

This is awesome because now we can calculate all kinds of interesting & usefull things:

Shear Stress:

$$\begin{aligned}\tau(x) &= \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} = \mu V_{\infty} \left. \frac{\partial \bar{u}}{\partial y} \right|_{y=0} \\ &= \mu V_{\infty} \left. \frac{\partial \bar{u}}{\partial \eta} \right|_{\eta=0} \cdot \left. \frac{\partial \eta}{\partial y} \right|_{y=0} \\ \tau(x) &= \mu V_{\infty} a_2 \sqrt{\frac{V_{\infty}}{xU}}\end{aligned}$$

Typically, you're used to seeing it in terms of a skin friction coefficient, C_f

$$C_{f,x} = \frac{\tau(x)}{\frac{1}{2} \rho V_{\infty}^2} = \frac{2a_2}{\sqrt{Re_x}} = \frac{0.664}{Re_x^{1/2}}, \quad \text{Reynolds number} \quad Re_x = \frac{\rho V_{\infty} x}{\mu}$$

↳ Skin friction coefficient for a flat plate in laminar flow.

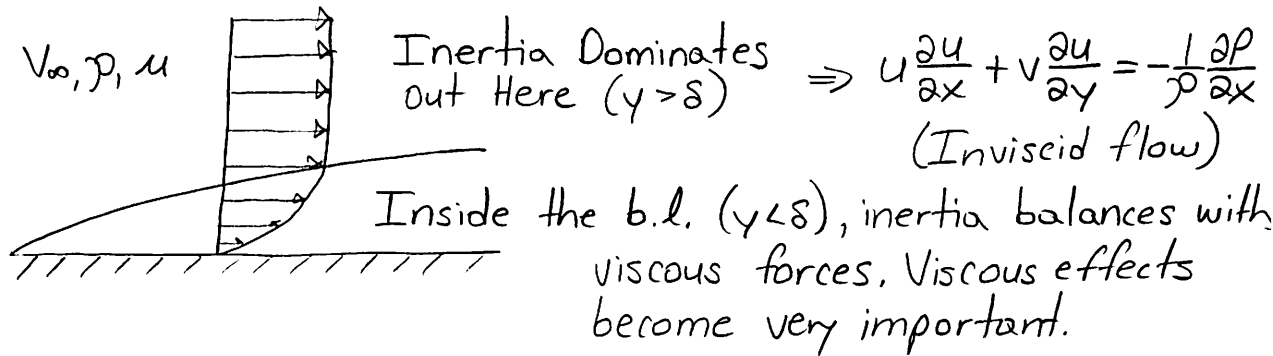
Typically we want the average shear:

$$\begin{aligned}\bar{\tau} &= \frac{1}{L} \int_0^L \tau(x) dx \Rightarrow \tau(x) = C \cdot \frac{1}{\sqrt{x}} \\ &= C \cdot \frac{1}{L} \int_0^L \frac{dx}{\sqrt{x}} = \frac{2C}{\sqrt{L}}\end{aligned}$$

$$C_0 = \frac{\bar{\tau}}{\frac{1}{2} \rho V_{\infty}^2} = \frac{1.328}{Re_L^{1/2}}, \quad Re_L = \frac{\rho V_{\infty} L}{\mu}$$

↳ Plate averaged skin friction coefficient

Note, there is a much easier way to solve for all this.
Use scaling:



Inertia $\sim$ Viscosity

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \sim \nu \frac{\partial^2 u}{\partial y^2}$$

$$V_{\infty} \frac{V_{\infty}}{L} + V_{\infty} \left(\frac{\delta}{L} \right) \cdot \frac{V_{\infty}}{\delta} \sim \nu \frac{V_{\infty}}{\delta^2}$$

$$\frac{V_{\infty}^2}{L} \sim \nu \frac{V_{\infty}}{\delta^2}$$

$$\delta^2 \sim \frac{\nu L}{V_{\infty}} \Rightarrow \delta \sim \sqrt{\frac{\nu L}{V_{\infty}}} \cdot \left(\frac{L}{L} \right) \Rightarrow \boxed{\delta \sim \frac{L}{\sqrt{Re_L}}} \text{ OMG! So easy!}$$

Note, this scaling result is great for giving you a sense of orders of magnitude, however not as accurate as analytical solutions.

Example | Calculate the boundary layer thickness on a 737 jet.

$$V_{\infty} = 400 \text{ miles/hour} = 177 \text{ m/s (Plane speed)}$$

$$\nu_{\text{air}} = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$$

$$L = 5 \text{ m (Length of the wing in the fuselage direction)}$$

$$Re_L = \frac{V_{\infty} L}{\nu} = 5.9 \times 10^7 \Rightarrow \boxed{\delta = \frac{5L}{Re_L} = 3.25 \text{ mm}} \Rightarrow \text{Less than 1cm thick!}$$

Very difficult to observe.